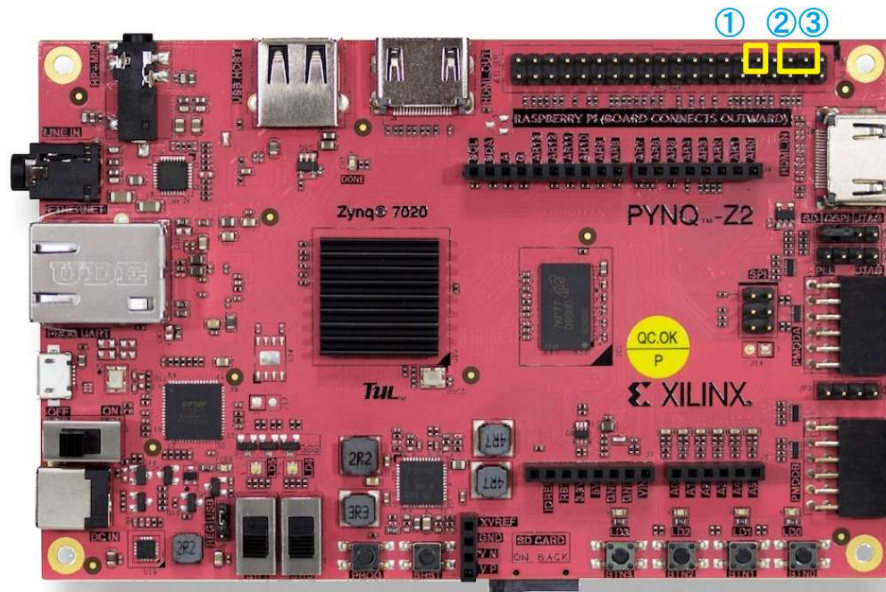


EE3220 TBL2

Connections between CH340 and PYNQ-Z2

1. Required wiring

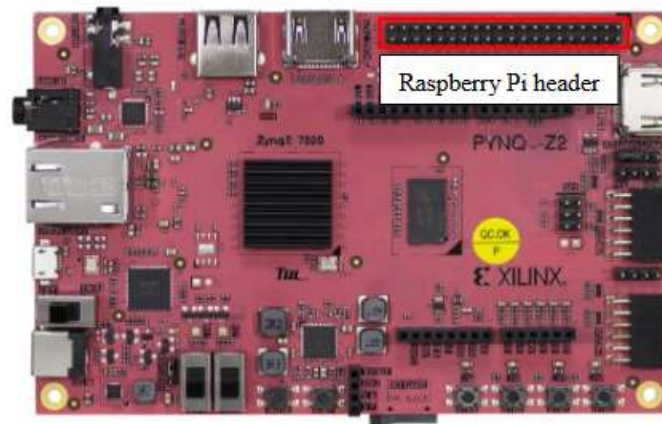
PYNQ-Z2 side	CH340 side	Direction	Notes
① GND	GND	/	Common ground is required for reliable UART signaling.
② UART_TX	RXD	PYNQ -> PC	TX and RX must be crossed.
③ UART_RX	TXD	PC -> PYNQ	Needed only if you want UART RX commands.
/	USB	CH340 -> PC	This is the serial link to PuTTY on PC.



2. Safety and handling notes

- Connect GND correctly. Without a common ground, UART levels may be unstable or fail completely. Missing GND usually causes unreliable communication rather than direct damage.
- In the default constraints file for TBL2, we are using pin W19 as uart_txd of PYNQ-Z2, W18 as uart_rxd of PYNQ-Z2.
- For the pin map of PYNQ-Z2, please see the following picture. **For the ground, please use the Ground pin between U7 and V6 according to the above (shown as ① in the above picture) and following picture. Please do not use the pin1 as Ground (the one to the left of W18) !!!**

The PYNQ-Z2 has a 40-pin Raspberry Pi connector with 28 pins connected to the Zynq device.



These pins can be used as GPIO, or to connect a standard Raspberry Pi peripheral.

G	W9	Y8	W8	Y7	Y6	Y16	G	W10	V10	V8	V	U8	V7	U7	G	V6	W19	W18	G
39	37	35	33	31	29	27	25	23	21	19	17	15	13	11	9	7	5	3	1
40	38	36	34	32	30	28	26	24	22	20	18	16	14	12	10	8	6	4	2
Y9	A20	B19	G	B20	G	Y17	F20	F19	U19	G	U18	W6	G	C20	Y19	Y18	G	V	V

G	Ground
V	3.3V
V	5V
...	Raspberry Pi header pin number
...	Zynq Pin

Table 22 Raspberry Pi header pin layout and Zynq PL pin assignments

- Do not connect CH340 3V3 / 5V to the PYNQ-Z2 UART pins. For this setup, only GND, TXD, and RXD are needed. Incorrect power wiring is a real damage risk.
- Use 3.3 V TTL-level UART signaling. If your CH340 board has a voltage-selection jumper or switch, set it to 3.3 V. (By default, the CH340 VCC pin and CH340 3V3 pin are already connected through jumper, see the following picture. Please do not change this connection.)



- Power off the board before rewiring. Hot-plugging jumpers onto adjacent pins can easily cause accidental shorts.
- Cross TX and RX. PYNQ UART_TX goes to CH340 RXD, and PYNQ UART_RX goes to CH340 TXD.

3. Sample graph

